

Pre-OS Boot

Sub-techniques (5)



Adversaries may abuse Pre-OS Boot mechanisms as a way to establish persistence on a system. During the booting process of a computer, firmware and various startup services are loaded before the operating system. These programs control flow of execution before the operating system takes control.^[1]

Adversaries may overwrite data in boot drivers or firmware such as BIOS (Basic Input/Output System) and The Unified Extensible Firmware Interface (UEFI) to persist on systems at a layer below the operating system. This can be particularly difficult to detect as malware at this level will not be detected by host software-based defenses.

ID: T1542

Sub-techniques: [T1542.001](#), [T1542.002](#), [T1542.003](#), [T1542.004](#), [T1542.005](#)

- Tactics: [Stealth](#), [Persistence](#)
- Platforms: Linux, Network Devices, Windows, macOS

Version: 2.0

Created: 13 November 2019

Last Modified: 12 May 2026

[Version Permalink](#)

Mitigations

ID	Mitigation	Description
M1047	Audit	Perform audits or scans of systems, permissions, insecure software, insecure configurations, etc. to identify potential weaknesses.
M1046	Boot Integrity	Use Trusted Platform Module technology and a secure or trusted boot process to prevent system integrity from being compromised. Check the integrity of the existing BIOS or EFI to determine if it is vulnerable to modification. ^[2] ^[3]
M1035	Limit Access to Resource Over Network	Prevent access to file shares, remote access to systems, unnecessary services. Mechanisms to limit access may include use of network concentrators, RDP gateways, etc.
M1026	Privileged Account Management	Ensure proper permissions are in place to help prevent adversary access to privileged accounts necessary to perform these actions
M1051	Update Software	Patch the BIOS and EFI as necessary.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0278	Detection Strategy for T1542 Pre-OS Boot	AN0774	Unusual modification of boot records (MBR, VBR) or EFI partitions not associated with legitimate patch cycles or OS upgrades. Registry or WMI events associated with firmware update tools executed from unexpected parent processes. API calls (e.g., DeviceIoControl) writing directly to raw disk sectors. Subsequent abnormal boot configuration changes followed by unsigned driver loads.
		AN0775	Detection of writes to /boot or EFI directories outside of expected package manager updates. Monitoring kernel log and auditd events for attempts to overwrite bootloader binaries (e.g., grub, shim). Unexpected execution of efibootmgr or dd writing to /dev/sdX devices followed by boot parameter changes.
		AN0776	Abnormal modification of EFI firmware binaries in /System/Library/CoreServices/ or NVRAM parameters not associated with OS updates. Unified logs capturing calls to bless or nvram commands executed from untrusted parent processes. Sudden unsigned next loads after EFI variable tampering.
		AN0777	Unexpected firmware image uploads via TFTP/FTP/SCP. Configuration changes modifying boot image pointers. Logs showing boot variable redirection to non-standard images. Anomalous reboots.

References

1. [Wikipedia. \(n.d.\). Booting. Retrieved November 13, 2019.](#)
2. [Trusted Computing Group. \(2008, April 29\). Trusted Platform Module \(TPM\) Summary. Retrieved June 8, 2016.](#)
3. [Microsoft. \(n.d.\). Secure the Windows 10 boot process. Retrieved April 23, 2020.](#)